FORMULE - SOLARNA ENERGIJA

δ - ugao deklinacije Sunca [°],
n - redni broj dana u godini,
L - geografska širina [°],
β - elevacioni ugao Sunca [°],
β_N - elevacioni ugao Sunca tokom solarnog podneva [°],
ω - satni ugao [°],
ϕ_S - azimutni ugao Sunca [°],
ϕ_c - azimutni ugao panela [°],
ST - solarno vrijeme [h],
LT - vrijeme na časovniku [h],
E - korekcija vremena [min],
Σ - ugao panela [°],
θ - ugao između svjetlosnog snopa i normale na panel [°],
Λ_{TM} - meridijan lokalnog vremena [°].

$$\delta = 23{,}45\, sin\left[\frac{360}{365}(n-81)\right], \qquad \beta_N = 90^\circ - L + \delta, \qquad sin\,\beta = cos\,L\, cos\,\delta\, cos\,\omega + sin\,L\, sin\,\delta,$$

$$sin\,\phi_S = \frac{cos\,\delta\, sin\,\omega}{cos\,\beta} \quad \text{ako je} \quad cos\,\omega \geq \frac{tg\delta}{tgL} \quad \text{onda je } |\phi_S| \leq 90^\circ, \quad \text{inače } |\phi_S| > 90^\circ, \omega = 15\left[\frac{^\circ}{h}\right]\cdot T_{dsp},$$

$$E = 9{,}87\, sin\,2B - 7{,}53\, cos\,B - 1{,}5\, sin\,B\ [min], \quad B = \frac{360}{364}(n-81), \ \omega = arccos(-tg\,L\, tg\,\delta)$$

$$ST = LT \pm 4\left[min/^\circ\right](\Lambda_{TM} - \Lambda_L) + E, \ cos\,\theta = cos\,\beta\, cos(\phi_S - \phi_C)\, sin\,\Sigma + sin\,\beta\, cos\,\Sigma.$$

G_0 - iradijansa van atmosfere [W/m^2],
m - odnos vazdušne mase,
G_B - iradijansa na normalnu podlogu [W/m^2],
A - prividni fluks zračenja van atmosfere,
k - optička debljina,
G_{BC} - direktna iradijansa na kolektor [W/m^2],
G_H - iradijansa na horizontalnu podlogu [W/m^2],
G_{DH} - difuzna iradijansa na horizontalnu podlogu [W/m^2],
G_{DC} - difuzna iradijansa na kolektor [W/m^2],
G_{RC} - reflektovana iradijansa na kolektor [W/m^2],
ρ - koeficijent refleksije,
G_C - iradijansa na kolektor [W/m^2].

$$G_0 = 1370\left[1 + 0{,}034\, cos\left(\frac{360n}{365}\right)\right], \ m = \frac{1}{sin(\beta)}, \ cos\,\theta = cos\,\beta\, cos(\phi_S - \phi_C)\, sin\,\Sigma + sin\,\beta\, cos\,\Sigma,$$

$$A = 1160 + 75\, sin\left[\frac{360}{365}(n-275)\right], \quad G_C = G_{BC} + G_{DC} + G_{RC}, \quad G_{RC} = \rho G_B(sin\,\beta + C)\left(\frac{1 - cos\,\Sigma}{2}\right),$$

$$G_{BH} = G_B\, cos(90^\circ - \beta) = G_B\, sin\,\beta, \qquad G_{DH} = C \cdot G_B, \ C = 0{,}095 + 0{,}04\, sin\left[\frac{360}{365}(n-100)\right],$$

$$I_B = Ae^{-km}, \ G_{DC} = G_{DH}\left(\frac{1 + cos\,\Sigma}{2}\right) = G_B C\left(\frac{1 + cos\,\Sigma}{2}\right), \qquad k = 0{,}174 + 0{,}035\, sin\left[\frac{360}{365}(n-100)\right],$$

$$G_{BC} = G_B\, cos(\theta),$$

$$G_C = Ae^{-km}\left[cos\,\beta\, cos(\phi_S - \phi_C)\, sin\,\Sigma + sin\,\beta\, cos\,\Sigma + C\left(\frac{1 + cos\,\Sigma}{2}\right) + \rho(sin\,\beta + C)\left(\frac{1 - cos\,\Sigma}{2}\right)\right].$$

Praćenje Sunca po dvije ose:

$$G_{BC} = G_B, \qquad G_{DC} = CG_B\left[\frac{1+cos\left(90^\circ-\beta\right)}{2}\right], \qquad G_{RC} = \rho\left(G_{BH}+G_{DH}\right)\left[\frac{1-cos\left(90^\circ-\beta\right)}{2}\right].$$

Praćenje Sunca po jednoj osi:

$$G_{BC} = G_B\, cos\,\delta, \quad G_{DC} = CG_B\left[\frac{1+cos\left(90^\circ-\beta+\delta\right)}{2}\right], \qquad G_{RC} = \rho\left(G_{BH}+G_{DH}\right)\left[\frac{1-cos\left(90^\circ-\beta+\delta\right)}{2}\right].$$

E_λ - spektralna snaga zračenja [W/(m^{2}μm)],
λ - talasna dužina [μm],
T - apsolutna temperatura [K],
E_f - energija jednog fotona [J],
f - frekvencija [Hz].
S - površina [m^2],
I_d - struja kroz diodu [A],
I_0 - inverzna struja zasićenja [A],
U_d - napon na diodi [V],
α_{TPV} - koeficijent promjene struje pri promjeni temperature ćelije,
G_C - iradijansa na PV modul,
$NOCT$ - normalna radna temperatura ćelije,
T_{pv}, T_{amb} - temperatura PV modula / ambijenta.

$$E_\lambda = \frac{3{,}74\cdot 10^8}{\lambda^5\left[e^{\frac{14400}{\lambda T}}-1\right]}, \quad \lambda_{max} = \frac{2898}{T[K]}, \quad E = S\sigma T^4, \quad E_f = hf = \frac{hc}{\lambda}, \quad I_d = I_0\left(e^{\frac{qU_d}{kT}}-1\right),$$

$$\sigma = 5{,}67\cdot 10^{-8}\ \frac{\mathrm{W}}{\mathrm{m^2K^4}}, \quad h = 6{,}626\cdot 10^{-34}\ \mathrm{Js}, \quad c = 3\cdot 10^8\ \frac{\mathrm{m}}{\mathrm{s}}, \quad q = 1{,}602\cdot 10^{-19}\ \mathrm{C}, \quad k = 1{,}38\cdot 10^{-23}\ \frac{\mathrm{J}}{\mathrm{K}},$$

$$T_{PV} = T_{amb} + \left(\frac{NOCT-20}{800}\right)G_C, \quad I_{KSPV} = I'_{KSPV}\left(1+\alpha_{TPV}\left(T_{PV}-T_{STC}\right)\right).$$

FORMULE - EOLSKA ENERGIJA

P_v - snaga vjetra [W],
A - površina koju zahvataju lopatice vjetroturbine [m^2],
v - brzina vjetra [m/s],
H - visina od tla [m],
z - dužina hrapavosti terena [m],
D - prečnik kružnice koji opisuju lopatice vjetroturbine [m],
η - efikasnost vjetroturbine,
$f(v)$ - gustina raspodjele vjerovatnoće.

$$P_v = \frac{1}{2}\rho A v^3, \qquad \rho = p_{15}K_T K_A = 1{,}225\cdot K_T K_A, \qquad \left(\frac{v}{v_0}\right) = \left(\frac{H}{H_0}\right)^\alpha, \qquad \left(\frac{v}{v_0}\right) = \frac{ln\frac{H}{z}}{ln\frac{H_0}{z}},$$

$$TSR = \frac{v_0}{v_v} = \frac{n\left[\frac{\mathrm{o}}{\mathrm{min}}\right]\frac{\pi D}{60}}{v_v}, \qquad \eta = \frac{1}{2}\left[\left(1+\frac{v_2}{v_1}\right)\left(1-\left(\frac{v_2}{v_1}\right)^2\right)\right], \qquad \left(v^3\right)_{sr} = \int_0^{+\infty} v^3 f(v)\,dv,$$

$$v_{sr} = \int_0^{+\infty} v f(v)\,dv, \qquad f(v) = \frac{k}{c}\left(\frac{v}{c}\right)^{k-1} e^{-\left(\frac{v}{c}\right)^k}, \qquad f(v) = \frac{\pi v}{2v_{sr}^2} e^{-\frac{\pi}{4}\left(\frac{v}{v_{sr}}\right)^2}, \qquad f(v) = \frac{2v}{c^2} e^{-\left(\frac{v}{c}\right)^2},$$

$$T\left(v_1 < v < v_2\right) = 8760\cdot\int_{v_1}^{v_2} f(v)\,dv, \qquad P\left(v_1 < v < v_2\right) = \int_{v_1}^{v_2} f(v)\,dv.$$